Re-deployment of AI solutions in Healthcare

The Research Conducted

By

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Under Supervision

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Participant Name (Optional):

Participant Designation:

Date:

Survey Instructions

Dear Participant

You are welcome to participate in this anonymous survey on "Re-deployment of AI solutions in healthcare". Participant's feedback will help us to understand how important it is to identify the differences between two environments when redeploying a Decision Support System (DSS). Thank you for taking the time to complete this survey and share your valuable insights.

Motivation

Development, testing, and validation of non-trivial DSS are typically laborious and costly. Hence, it is desirable to re-deploy such solutions across all health care providers whenever possible.

A significant barrier to smooth re-deployment is high variability in healthcare facilities, data, and clinical practice. To mitigate the impact of this barrier, we propose a framework supporting DSS re-deployment. The first step is identifying the differences in the input variables set necessary to drive the source DSS and the variables available in the destination healthcare facility data ecosystem.

Key Instructions

- Completing the survey should take about 20 minutes. We genuinely value your perspective and would be grateful for your thoughts on each question, as this will enrich our research. Please respond only to the questions you feel comfortable answering—you're always welcome to skip any question.

1. Where do you work? [tick one]

   ○ Australia, If yes, please select a State

   Choose an item.

   ○ Overseas, please enter country

   [ ]

2. What is your education? [tick all applicable answers]

   ☐ Clinical (allied health, nursing, pharmacy, medical, etc.)
   ☐ Information Technology (specialisation)
   ☐ Information Systems (specialisation)
   ☐ Other (text box):

   [ ]

3. What is your current principal occupation? [tick the closest option]
   - ☐ Clinical (enter details such as nursing, pharmacy, allied health, general practice) text here:
   - ☐ Health informatics
   - ☐ Clinical informatics
   - ☐ Other IT practitioner (text box):

4. What is your employment area? [tick all that apply]
   - ☐ Academia
   - ☐ Health care provider
   - ☐ Health-related government/public service
   - ☐ Health-related industry
   - ☐ IT solutions provider
   - ☐ Consultancy company
   - ☐ Other (text box):

5. What is your experience with DSS? [tick one]
   - ☐ 1 year.
   - ☐ 2-5 years.
   - ☐ 5-10 years.
   - ☐ 10+ years.

6. Have you ever been involved in any stage of deploying or re-deploying a DSS, such as initial business requirement gathering, preparing software requirement specifications or functional documents, designing implementation approaches, developing the system architecture, or managing the deployment process? [Tick one, if tick on "yes"; sub-options can have one or more answers]
   - ☐ No
   - ☐ Yes
     - ☐ Gathering of Business requirements
     - ☐ Preparing Functional and software document and Technical Feasibility
     - ☐ Wireframing and prototyping based on Functional document
     - ☐ Devising the overall architecture and workflow
     - ☐ Devising and development of adaptation strategies
     - ☐ Minimal Viable Product (MVP) stage
     - ☐ Enterprise level development
     - ☐ Deployment and testing of overall solution
     - ☐ Other (text box):

7. What do you think is the motivation of DSS portability in healthcare? [Rank each option from 1-5; 1 being highest priority; two factors can have the same priority]

- To save money
- To keep updated knowledge bases
- To reduce manual effort
- To increase AI adoption in healthcare
- Enables continuity of care and consistent decision support for patients
- Other:

8. A lot of applications are now published to support decision-making in health. What are the major barriers in re-deployment of a DSS [prioritize the barriers 1-5 – 1 being most important; several items can have the same priority]

- Differences in health practice
- Differences in patient mix
- Differences in available data
- Cost of adoption
- Lack of capability (lack of expertise in re-implementation of a DSS)
- Lack of capacity (not enough people to do it)
- Lack of interest (DSS do not offer benefits worth the effort/cost)
- Other (text box):

9. Is there a need for a framework facilitating re-deployment of CDSS across healthcare providers? [choose one answer best expressing your opinion]

- ○ Strongly disagree
- ○ disagree
- ○ neutral
- ○ agree
- ○ strong agree
- ◉ other (text box)

General Case

Each healthcare provider is a specific data ecosystem containing different variables, rules, regulations, and patient demographics. Decision Support System was designed and tested as part of a specific ecosystem with a defined set of variables as input. Porting a piece of Software/Decision Support System from one healthcare provider to another is facing several complexities (issues). In practice, there are several key questions (heuristics) you must consider before porting a DSS from one healthcare provider to another which are following below:

Variable Availability and Access

- Are all the variables that the software requires present in the target ecosystem?
- Even if a variable exists, is it accessible (e.g., stored in the same database, within same governance, or visible to the DSS)?

Definition Consistency

- When you match source vs. target variables based on name or description, do they really mean the same thing?

Measurement Method and Units

- Do both ecosystems use the same unit of measurement (e.g., mg/dL vs. mmol/L)?
- Is the method of capture identical (For example, "Blood Pressure" measured according to the WHO standards by a trained nurse using a floating zero, calibrated sphygmomanometer – vs. patient self-measured blood pressure at home conditions using a non-calibrated inexpensive automatic device)?
- Which device was used, and does that introduce any systematic bias?

Metadata and Reference Ranges

- Does the target system store the same metadata (e.g., ranges, etc.)?
- Are the reference or allowable ranges for each variable consistent, or do they need adjustment?

Data Distribution and Patient-Mix Differences

- Even if a variable exists with the same name and unit, does its distribution differ because of population heterogeneity or different clinical practices?
- For example, average Body Mass Index (BMI) may be higher in one region, or lab values can shift due to differing patient demographics; that changes could affect the DSS's performance.

Scenario:

Several DSSs (and regular updates) need to be re-deployed from Hospital 1 to Hospital 2. A typical DSS requires 50 variables as input, which are well described in the DSS documentation.

10. Given the scenario – would you: [choose one answer best expressing your opinion]

    - ○ Prefer expert-driven, manual adaptation process
    - ○ Prefer semi-automatic adaptation with human-in-the-loop
    - ○ Accept fully automatic adaptation without human involvement

11. Is adopting a reference terminology standard important for the re-deployment task? [choose one answer best expressing your opinion]

    - ○ Unimportant
    - ○ Somewhat unimportant
    - ○ Somewhat important
    - ○ Important
    - ○ Very important
    - ○ I do not have a strong opinion

12. Which terminology standards do you think are important to support DSS re-deployment? [Rank each option from 1-5; 1 being highest priority; two factors can have the same priority]:

    - HL7 FHIR (Fast Healthcare Interoperability Resources) ☐
    - ICD-9/10 (The International Classification of Diseases -I9/I10) ☐
    - SNOMED-CT (Systematized Nomenclature of Medicine - Clinical Terms) ☐
    - LOINC (Logical Observation Identifiers Names and Codes) ☐
    - UMLS (Unified Medical Language System) ☐
    - Other: several entries can be added here (text box):

      ☐ ☐

13. Is it important to capture and compare the **definitions** of the terms in the selected terminology? [choose one answer best expressing your opinion]

- ○ Unimportant
- ○ Somewhat unimportant
- ○ Somewhat important
- ○ Important
- ○ Very important
- ○ No opinion/not applicable/Comment (text box):

14. What contextual information would you consider? [Rank each option from 1-5; 1 being highest priority; two factors can have the same priority]

- Units of measure
- Reference intervals
- Circumstances of measurement
- Other (text box):

15. What kind of metadata do you think is important for porting a DSS?
(text box):

16. Are differences in patient case mix important in DSS re-deployment? [choose one answer best expressing your opinion]

- ○ Unimportant
- ○ Somewhat unimportant
- ○ Somewhat important
- ○ Important
- ○ Very important

17. What is the best relative importance of the dimensions of comparison? [Rank each option from 1-5; 1 being highest priority; two factors can have the same priority]

- To check the label match with respect to the terminology
- To check the definition of the term with respect to the ontology
- To check the way of representation of the value
- To check the same interpretation of associated data
- To check the distribution of value in case of patient mix

Concluding questions:-

18. Are you available for a follow-up interview? [tick one]

    - ◯ Yes
    - ◯ No
    - ◯ May be

19. Can you please recommend other potential participants for this research [Please write the potential participant's name and email address]:

    Text here:

Any Suggestions: